

Particulate Matter

Monitoring & Health Effects

Issue 2026-10-05 | Entry window 5 October 2026 (full day) | 13 records screened in scope

13

RECORDS IN SCOPE
(39 EXCLUDED)

7

OF 10 SUBTOPIC
BANDS POPULATED

3

EFFECT ESTIMATES
WITH A CI

2

TIER-A
RECORDS

5

RECORDS CARRIED ON
METADATA ALONE

Signal of the day

2.1 million adults, 19 million person-years, and the exposure that carries the dementia signal is the one no low-cost node measures.

Batisse and colleagues (*Epidemiology*, tier A) linked high-resolution residential **ultrafine particle number** and **UFP size** to cause-of-death records for 2.1 million adults in Montreal and Toronto, 2001–2019: **20,560** neurodegenerative deaths, and per **10,000 particles cm⁻³** a dementia-mortality **HR of 1.22** (1.17–1.27) — stronger after adjusting for particle size, positive across every dementia subtype, largest for vascular dementia. **Caveat:** UFP number and NO₂ share the same traffic gradient, so co-pollutant adjustment cannot cleanly separate them, and dementia is under-ascribed on death certificates. **Why it matters for sensing:** UFP is unregulated, and an optical low-cost sensor is blind to it — it reports a scattering proxy for mass, and sub-100 nm particles scatter almost nothing. A network built to track PM_{2.5} mass reports approximately zero information about the exposure this cohort implicates. The same day, a tethered-balloon synthesis over Houston shows the other blind spot: profiles that vary severely with boundary-layer depth above a surface node that cannot see any of it.

What changed today

- ▶ **Gene–environment interaction in 313,534 people.** UK Biobank: PM_{2.5} **HR 1.22 per 5 μg m⁻³** (1.19–1.26) across five chronic-disease groups, with additive interaction between high polygenic risk and high PM_{2.5} for neurodegenerative disease (RERI 0.14) and diabetes (0.21). A healthy-volunteer cohort over a narrow exposure range, five outcomes, four pollutants, and no multiplicity correction on the interaction terms.
- ▶ **The 1.5 m reading is not the 20 m reading.** A three-level mast inside a Shenyang urban forest through the heating season found PM_{2.5} and PM₁₀ *higher* at 10 and 20 m than at pedestrian level, O₃ and CO higher at 1.5 m, NO₂ and SO₂ peaking mid-canopy. One site, one winter — but network siting guidance assumes a well-mixed sub-canopy layer, and this is direct evidence against it.
- ▶ **Triangulation that is not independent.** Pretoria: hazard quotients above unity in all age groups, case-crossover **OR 1.027** (1.006–1.049) per 10 μg m⁻³, and 158–748 admissions avertable depending on the counterfactual. The three methods re-use one exposure series, and summing element-specific attributable cases double-counts the mass those elements are part of.
- ▶ **Respirable dust is the wrong metric for a silica hazard.** Personal sampling across five stone-working tasks in Thailand: all means below the Thai OEL and TLV, highest in carving (0.051 mg m⁻³), modelled ILCR 3.3×10^{-7} . No crystalline silica fraction is reported, and an EPA inhalation-unit-risk framework applied to mixed mineral dust yields a number with no exposure–response behind it.

Window continuity

The 4 October issue closed with `last_entry_date` at 4 October. This issue collects **5 October in full**, continuous with it. Harvest and screening were completed on the 6 October run and cached; this run authored from that cache without re-querying. Dailies for 6–8 October and the W40 weekly (27 September–3 October) remain owed and follow in this backfill.

Corpus analytics

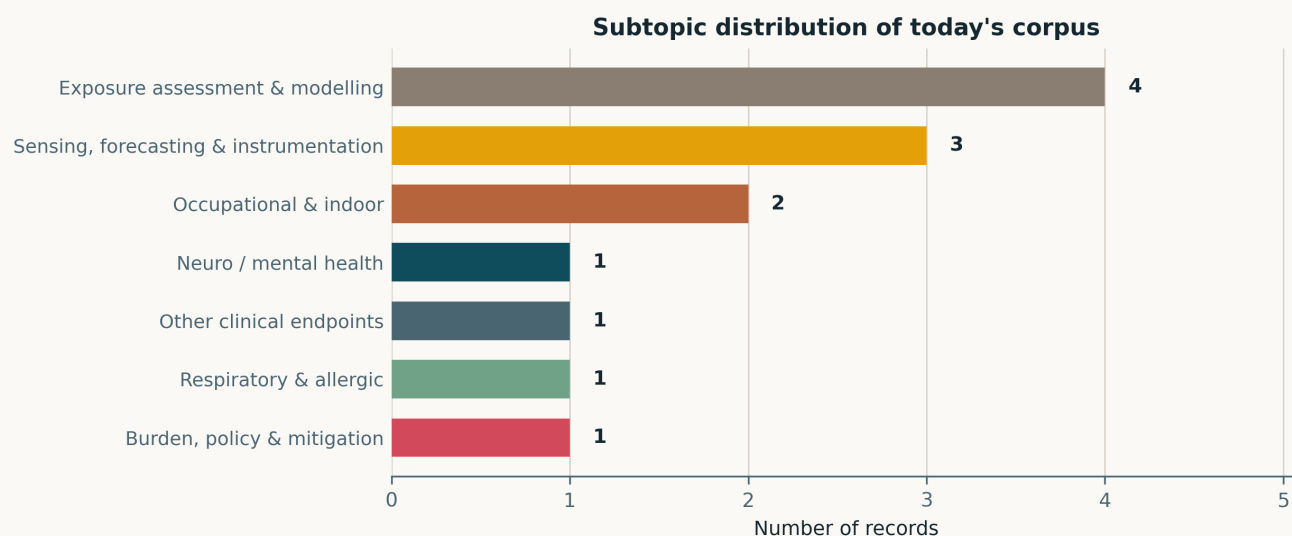


Fig. Subtopic assignment of the 13 in-scope records. **Seven of ten bands populated**; *Exposure assessment* leads with **four** (three of them metadata-only), *Sensing* has three, *Occupational & indoor* two, and one each for *Neuro*, *Other clinical*, *Respiratory* and *Burden*. The clinical side is thin but carries both tier-A records.

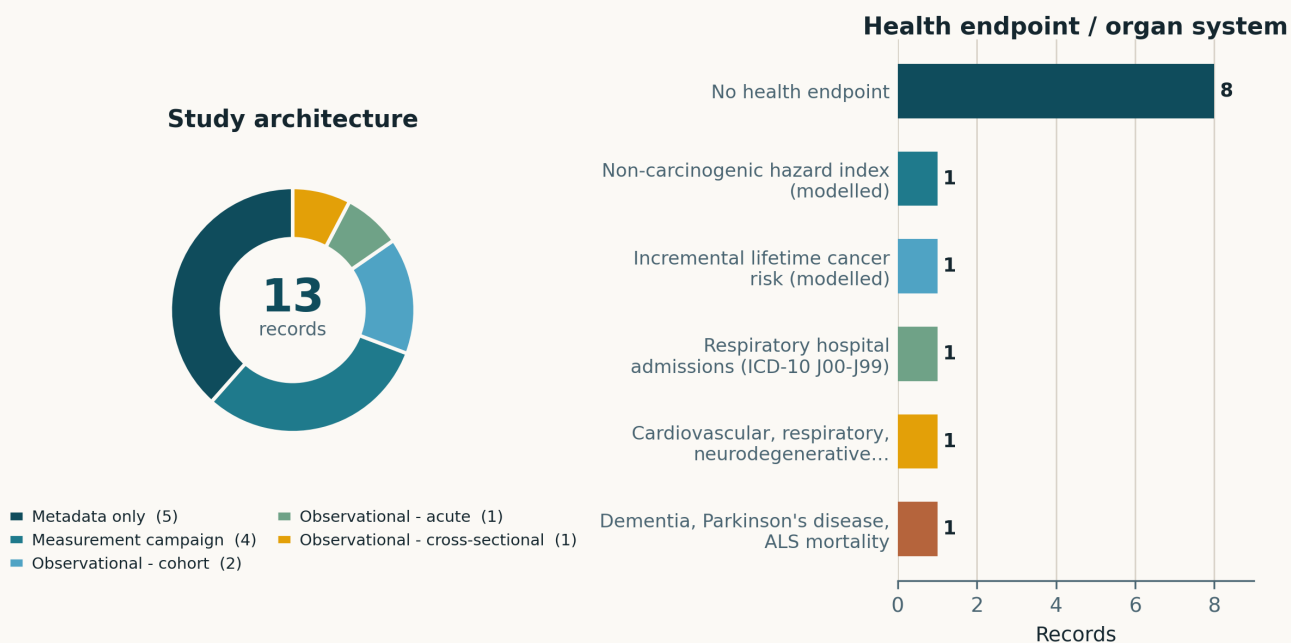


Fig. Left — architecture: **five** records carry no abstract and assert no design; **four** are measurement campaigns (the canopy mast, the tethered balloon, the Nagpur monitoring year and the Xinxiang mercury sampling); two are cohorts (CanCHEC, UK Biobank), one acute (the case-crossover) and one cross-sectional (the stone carvers). Right — the majority of records report no health endpoint; the four that do are split between neurodegenerative mortality, multi-system chronic disease, respiratory admissions and a modelled cancer risk.

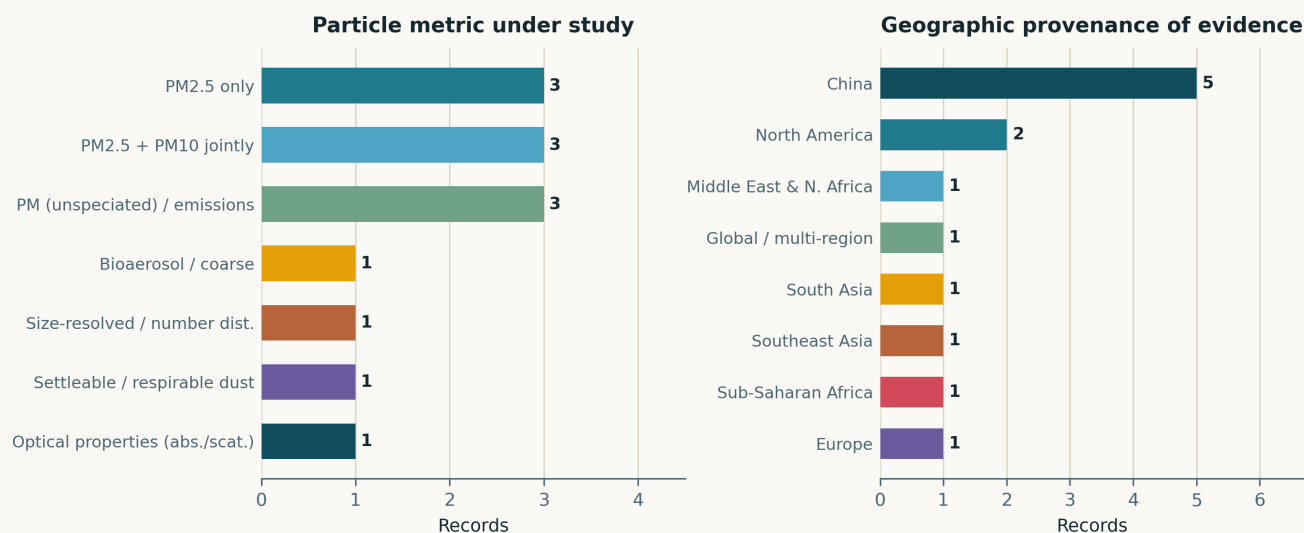


Fig. Left — particle metric: unspicated PM and PM_{2.5} + PM₁₀ jointly and PM_{2.5} only take three each, with one record each for size-resolved number distribution (the Houston profiles), optical properties, settleable/respirable dust (the personal sampling) and bioaerosol. The UFP cohort is the only record whose exposure is a *number* concentration. Right — geography: China five, North America two, and one each for Europe, Sub-Saharan Africa, South Asia, Southeast Asia, the Middle East & North Africa and global / multi-region.

Life-course windows addressed today (records may span several stages)

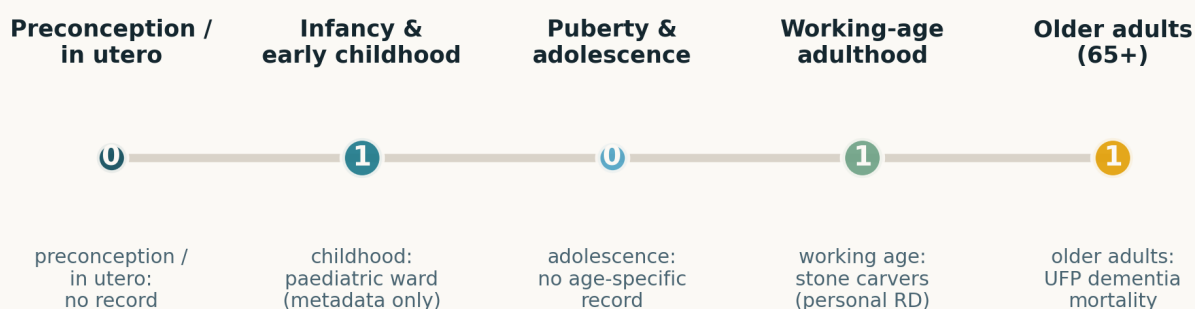


Fig. Life-course windows: older adults one (the UFP dementia-mortality cohort), working age one (stone carvers under personal sampling), childhood one (the paediatric-ward bioaerosol model, metadata only). In utero and adolescence have no record. The UK Biobank cohort spans adulthood and is not assigned to a single window.

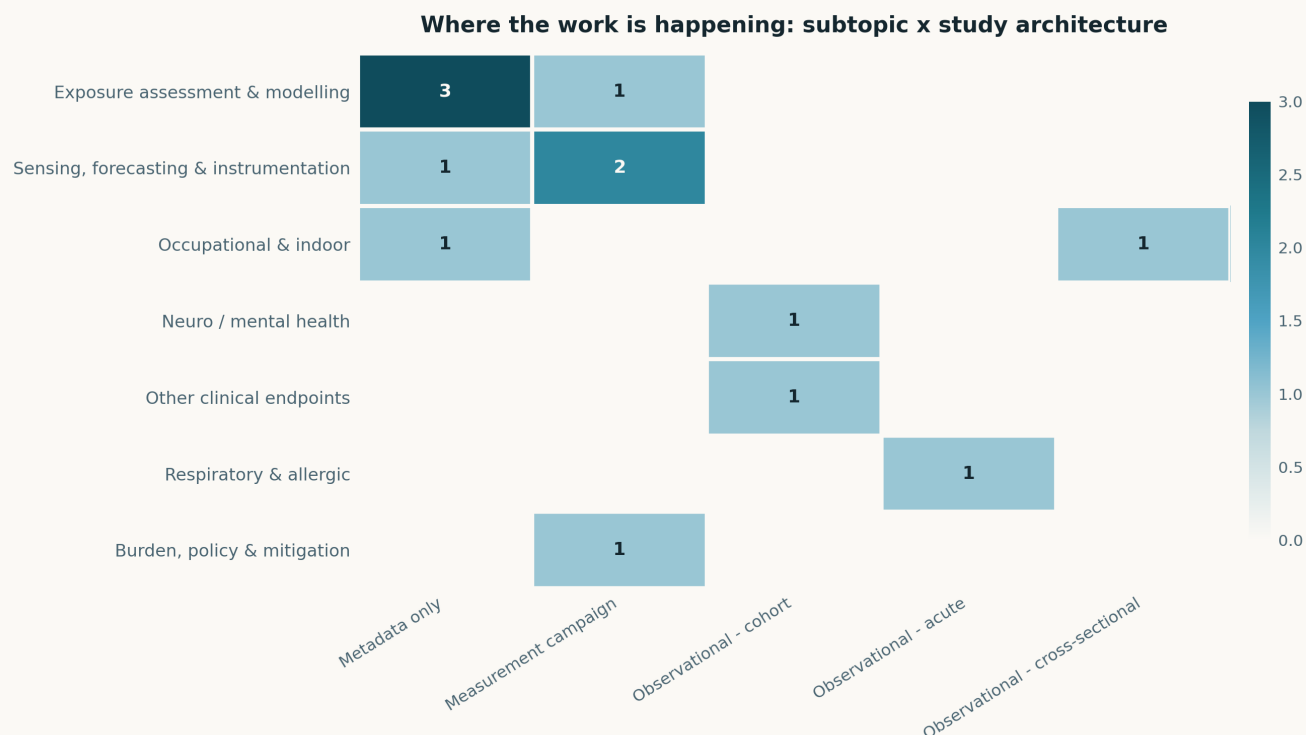


Fig. Subtopic × study architecture for the 13 records. The metadata-only block spans exposure assessment, sensing and occupational; measurement campaigns occupy sensing, exposure and burden; the three epidemiological architectures each sit alone in their clinical band.

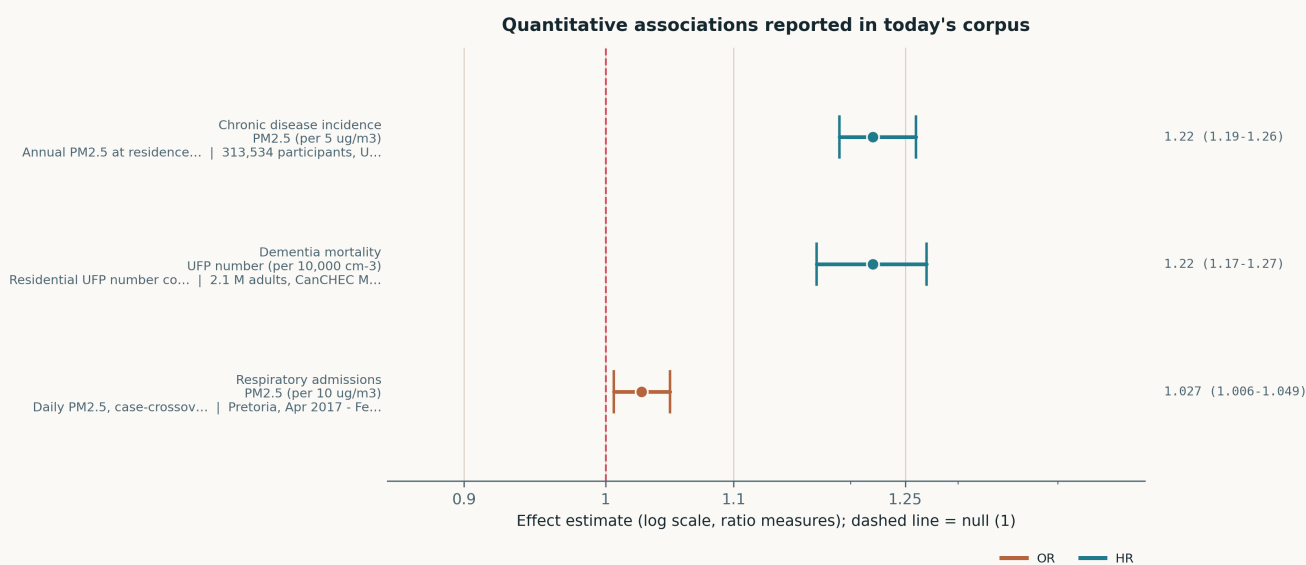


Fig. Three estimates with 95% CIs, log scale, from three different cohorts and three different exposure contrasts — per 10,000 particles cm⁻³ of UFP number (dementia mortality), per 5 µg m⁻³ of annual PM_{2.5} (chronic-disease incidence) and per 10 µg m⁻³ of daily PM_{2.5} (respiratory admissions). The first two agree at 1.22 by coincidence of scaling, not of effect: one is a chronic number-concentration contrast and the other an annual-mass contrast, and they are not poolable. The third is an acute estimate and belongs to a different estimand entirely.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health

1 record

Batisse et al. 2026 Epidemiology

Long-term exposure to ultrafine particles and mortality from neurodegenerative diseases: a population-based cohort study

2.1 million adults in the Canadian Census Health and Environment Cohorts, 2001-2019, 19 million person-years and 20,560 neurodegenerative deaths. Per 10,000 particles/cm³ of UFP number, dementia mortality HR 1.22 (95% CI 1.17-1.27), strengthened after adjustment for UFP size; positive for every dementia subtype, largest for vascular dementia. Models were run with and without inverse-probability censoring weights for competing events and adjusted for co-pollutants. Limitation: postal-code exposure from a land-use-regression UFP surface, no mobility or indoor time, and cause of death on the death certificate under-ascribes dementia; UFP number and NO₂ share the traffic gradient, so co-pollutant adjustment cannot fully separate them. Relevance: high - UFP is unregulated and unmeasured by every optical low-cost node, which sees mass, not number; this is the strongest current argument for CPC-class instruments in dense networks.

doi: 10.1097/ede.0000000000002054

Respiratory & allergic

1 record

Howlett-Downing and Wichmann 2026 Int J Environ Health Res

Triangulating health risks of PM_{2.5} and trace elements in Pretoria: evidence from health risk assessment, case-crossover, and concentration-response function analyses

Three methods applied to the same secondary dataset: hazard quotients, a case-crossover analysis and concentration-response functions. Hazard quotients exceeded unity in adults, children and infants; the case-crossover OR was 1.027 (95% CI 1.006-1.049) per 10 µg m⁻³ PM_{2.5}. Meeting the South African standard would avert 158 admissions annually, rising to 342 and 748 under the WHO daily and annual guidelines; element-specific functions give 110-608. Limitation: the three methods are not independent evidence - they re-use one exposure series from a small number of sampling days, and summing element-specific attributable cases double-counts the mass they are part of; hazard quotients are screening-level and not comparable to epidemiological risk. Relevance: moderate - Southern African monitoring is thin, and a saturation low-cost network would resolve the within-city contrast this design cannot.

doi: 10.1080/09603123.2026.2733112

Exposure assessment & modelling

4 records

Chavan and Lataye 2026 Environ Sci Pollut Res Int

Application of Openair package in R programming and PCA: unveiling air quality dynamics and meteorological influences

A full year of regulatory monitoring data for a tier-2 Indian city analysed with openair and PCA. Winter AQI reached 492 (severe) in February with monsoon improvement; varimax PCA gave two components explaining 73.1% of variance, read as anthropogenic combustion (PM, NO₂) and photochemistry (O₃, temperature); PM_{2.5} and PM₁₀ exceeded CPCB standards on 27-29 days per winter month. Limitation: descriptive reanalysis of one station-year with no source apportionment, no receptor model and no validation - PCA components here are correlation structure, not sources, and the policy recommendations do not follow from the analysis. Relevance: low as method, moderate as a reminder that tier-2 cities have one regulatory station and no spatial information at all.

doi: 10.1007/s11356-026-38275-w

Zhao et al. 2026c Atmos Pollut Res

Spatial Heterogeneity and Multi-scale Statistical Associations of Urban Air Pollution Revealed by Interpretable Machine Learning in Shenzhen, China

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 5 Oct with no abstract. By title, an interpretable machine-learning treatment of intra-urban spatial heterogeneity at several scales in Shenzhen. Scale dependence of the predictor set is the question that decides how dense a network has to be before adding nodes stops buying information, so the result matters more than the method. Relevance: potentially high; on the retry list.

doi: 10.1016/j.j.apr.2026.103230

Hopke et al. 2026 Atmos Pollut Res

Characterization of PM Pollution Episodes in Kuwait

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 5 Oct with no abstract. By title, a characterisation of PM pollution episodes in Kuwait. Dust-dominated Gulf environments are where optical low-cost sensors fail hardest - coarse mineral dust sits outside the calibrated size and refractive-index range of nephelometric nodes - and the episode statistics are the input any regional network design needs. Relevance: potentially high; on the retry list.

doi: 10.1016/j.j.apr.2026.103226

Shi et al. 2026 Environ Pollut

Compositional transition and coupled formation pathways of PM_{2.5} pollution in a typical industrial city: An empirical study based on Handan, China

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, an empirical account of how PM_{2.5} composition in a heavily industrial Hebei city has shifted and how its formation pathways couple. A compositional transition from primary to secondary is the slow change that silently alters the relationship between an optical sensor's scattering signal and gravimetric mass over years of a network's life. Relevance: moderate; on the retry list.

doi: 10.1016/j.envpol.2026.129296

Sensing, forecasting & instrumentation

3 records

Chang et al. 2026 Environ Pollut*Wintertime vertical stratification of air pollutants and meteorological associations within an urban forest canopy in Shenyang, China*

A three-level monitoring mast inside an urban forest in a cold-climate northern Chinese city through the winter heating season. PM_{2.5} and PM₁₀ were generally higher at 10 and 20 m than at pedestrian level, O₃ and CO higher at 1.5 m, and NO₂ and SO₂ peaked mid-canopy; relative humidity and wind speed dominated the statistical and machine-learning associations, but height-dependently. Limitation: one site, one winter, no replication mast and no flux measurement, so the inversion of the usual near-ground maximum is described rather than explained - and at wintertime humidity, uncorrected optical PM readings are themselves height-correlated through the humidity profile. Relevance: high - siting guidance for low-cost networks assumes a well-mixed layer below canopy height; this is direct evidence that the 1.5 m reading is not the 20 m reading.

doi: 10.1016/j.envpol.2026.129270

Mei et al. 2026 Atmos Chem Phys*Synthesis of the tethered balloon system and other TRACER campaign measurements elucidates aerosol property profiles*

149 tethered-balloon flights over greater Houston in summer during the DOE ARM TRACER campaign, with back-trajectory k-means clustering into marine, mixed and urban/long-range air masses. Profiles vary strongly with cluster, modulated by boundary-layer depth and coastal circulation; the marine cluster had CCN at 0.8% supersaturation below 1,000 cm⁻³, urban and mixed clusters higher. Limitation: one coastal city, one summer, and CCN is inferred from size distribution under an assumed kappa rather than measured, so composition heterogeneity propagates straight into the activated fraction; tethered-balloon sampling is intermittent and fair-weather-biased. Relevance: high - this is the measurement that exposes the central fiction of a surface network, namely that a 2 m reading represents the column a satellite retrieval sees.

doi: 10.5194/acp-26-13885-2026

Hazelwood et al. 2026 J Aerosol Sci*Size-Resolved Particle Dry Deposition Velocities in a Controlled Spherical Chamber: Effects of Turbulence, Surface Complexity, and Particle Physicochemical Properties*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, a chamber determination of size-resolved dry deposition velocity against turbulence intensity, surface complexity and particle physicochemical properties. Deposition velocity is the term that sets indoor particle loss rates and therefore the inferred air-change rate in every filtration study that does not measure one, and it is the largest uncertainty in indoor-outdoor infiltration models. Relevance: potentially high; on the retry list.

doi: 10.1016/j.jaerosci.2026.106903

Occupational & indoor

2 records

Ketsakorn and Chaiyadej 2026 PLoS One*Probabilistic health risk assessment of respirable dust exposure among stone carvers in Thailand using average daily dose and Monte Carlo simulation*

Personal sampling across mining, cutting, splitting, knocking and carving, with a 10,000-iteration Monte Carlo ILCR using US EPA exposure parameters. Mean respirable dust was below both the Thai OEL and the TLV everywhere, highest in carving at 0.051 mg/m³; ILCR stayed under 1e-6, highest in carving at 3.3e-7. Limitation: respirable dust mass is the wrong metric for a silica hazard - no crystalline silica fraction is reported, and an EPA inhalation-unit-risk framework applied to a mixed mineral dust produces a number with no established exposure-response behind it; small sample, one season. Relevance: low-moderate - the task-level contrast is exactly what a wearable logging monitor resolves and a single shift-average sample does not.

doi: 10.1371/journal.pone.0359706

Zhao et al. 2026d Build Environ*Behavior-resolved bioaerosol exposure modeling reveals scale-dependent occupational risk in pediatric wards*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, a bioaerosol exposure model in paediatric wards resolved by occupant behaviour, reporting scale-dependent occupational risk. Behaviour-resolved models are the class of work that decides whether a fixed room monitor can stand in for a worker's exposure at all, and the paediatric ward is a setting where that substitution is routinely assumed. Relevance: potentially high; on the retry list.

doi: 10.1016/j.buildenv.2026.115364

Burden, policy & mitigation

1 record

Zhao et al. 2026b Atmosphere*Pollution Characteristics, Influencing Factors and Health Risk Assessments of Mercury in Atmospheric PM_{2.5} from Xinxiang, China*

Year-long filter sampling gives an annual mean PM_{2.5}-bound Hg of 0.399 ng/m³, significantly higher in winter (p<0.001) and in the heating than the non-heating period, positively correlated with SO₂, CO, PM_{2.5}, OC, EC, Cl-, Zn and Ca and negatively with temperature, O₃ and V - a coal-combustion signature. Hazard index 2.06e-3 (children) and 8.46e-4 (adults), both far below 1, with ingestion the dominant modelled pathway. Limitation: the samples are from 2015 and describe an emissions regime two major control phases out of date; an ingestion-dominant pathway for an airborne metal is an artefact of the EPA soil-oriented default parameters. Relevance: low - logged as a composition time point, not as a risk estimate.

doi: 10.3390/atmos17100974

Other clinical endpoints

1 record

Tian et al. 2026 Hum Genet*Annual average exposure to ambient air pollution, polygenic risk, and incidence of major chronic diseases: a large-scale prospective cohort study*

313,534 UK Biobank participants with LUR annual pollutant estimates and weighted polygenic risk scores for each of five disease groups; Cox models with RERI and AP for additive interaction. PM_{2.5} carried the strongest association, HR 1.22 per 5 $\mu\text{g m}^{-3}$ (95% CI 1.19-1.26); high genetic risk acted independently (diabetes HR 1.77 per SD of PRS), with significant additive interaction between high PRS and high PM_{2.5} for neurodegenerative disease (RERI 0.14) and diabetes (RERI 0.21). Limitation: UK Biobank is a healthy-volunteer cohort with a narrow PM_{2.5} range, the LUR surface is a single baseline-era estimate carried forward, and five outcomes with four pollutants invites selective emphasis - the interaction terms are modest and not corrected for multiplicity. Relevance: moderate - gene-environment interaction raises the value of resolving exposure well at the individual level, which is the case for personal rather than ambient monitoring.

doi: 10.1007/s00439-026-02875-w

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Batisse et al.	Long-term exposure to ultrafine particles and mortality from neurodegenerative diseases: a population-based cohort study	<i>Epidemiology</i>	Retrospective cohort	High-resolution UFP number concentration and UFP size at residential postal code, 3-year moving average	Canada (Montreal and Toronto)
Respiratory & allergic					
Howlett-Downing and Wichmann	Triangulating health risks of PM _{2.5} and trace elements in Pretoria: evidence from health risk assessment, case-crossover, and concentration-response function analyses	<i>Int J Environ Health Res</i>	Time-stratified case-crossover	Sampled PM _{2.5} mass and PM _{2.5} -bound trace elements, Apr 2017 - Feb 2020	South Africa (Pretoria)
Occupational & indoor					
Ketsakorn and Chaityadej	Probabilistic health risk assessment of respirable dust exposure among stone carvers in Thailand using average daily dose and Monte Carlo simulation	<i>PLoS One</i>	Observational - cross-sectional	Personal respirable dust, NIOSH 7601, 8-h TWA across five task groups	Thailand
Zhao et al.	Behavior-resolved bioaerosol exposure modeling reveals scale-dependent occupational risk in pediatric wards	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - behaviour-resolved bioaerosol exposure model in hospital wards (by title)	China (by journal and author affiliation; not asserted)
Sensing, forecasting & instrumentation					
Chang et al.	Wintertime vertical stratification of air pollutants and meteorological associations within an urban forest canopy in Shenyang, China	<i>Environ Pollut</i>	Long-term field measurement campaign	PM _{2.5} , PM ₁₀ with O ₃ , NO ₂ , SO ₂ , CO and meteorology at 1.5, 10 and 20 m	China (Shenyang)
Hazelwood et al.	Size-Resolved Particle Dry Deposition Velocities in a Controlled Spherical Chamber: Effects of Turbulence, Surface Complexity, and Particle Physicochemical Properties	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only - size-resolved dry deposition velocity in a controlled chamber (by title)	Not recoverable from metadata
Mei et al.	Synthesis of the tethered balloon system and other TRACER campaign measurements elucidates aerosol property profiles	<i>Atmos Chem Phys</i>	Long-term field measurement campaign	Vertical profiles of aerosol number concentration and size distribution; CCN inferred by kappa-Koehler	USA (Houston, Texas)
Exposure assessment & modelling					
Chavan and Lataye	Application of Openair package in R programming and PCA: unveiling air quality dynamics and meteorological influences	<i>Environ Sci Pollut Res Int</i>	Monitoring-record analysis	2023 CAAQMS records: PM ₁₀ , PM _{2.5} with six co-pollutants and four meteorological variables	India (Nagpur)
Hopke et al.	Characterization of PM Pollution Episodes in Kuwait	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - PM episode characterisation (by title)	Kuwait
Shi et al.	Compositional transition and coupled formation pathways of PM _{2.5} pollution in a typical industrial city: An empirical study based on Handan, China	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} compositional transition and coupled formation pathways (by title)	China (Handan, Hebei)
Zhao et al.	Spatial Heterogeneity and Multi-scale Statistical Associations of Urban Air Pollution Revealed by Interpretable Machine Learning in Shenzhen, China	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - multi-scale spatial heterogeneity via interpretable ML (by title)	China (Shenzhen)

First author	Focus	Journal	Design	Metric	Region
Burden, policy & mitigation					
Zhao et al.	Pollution Characteristics, Influencing Factors and Health Risk Assessments of Mercury in Atmospheric PM _{2.5} from Xinxiang, China	<i>Atmosphere</i>	Source apportionment	PM _{2.5} -bound mercury with OC/EC, water-soluble ions and trace elements, full-year 2015 sampling	China (Xinxiang, Henan)
Other clinical endpoints					
Tian et al.	Annual average exposure to ambient air pollution, polygenic risk, and incidence of major chronic diseases: a large-scale prospective cohort study	<i>Hum Genet</i>	Prospective cohort	Annual PM _{2.5} , PM ₁₀ , NO ₂ , NO _x at residence from land-use regression	United Kingdom (UK Biobank)

Provenance and method

Entry window **5 October 2026**, single day, continuous with the 4 October issue. Harvest and screening for this day were completed on the **6 October** run and cached (`cache/2026-10-05/_fresh.json`); this run authored from that cache without re-querying. **52** fresh candidates after cross-day deduplication, **13 admitted**, **39 rejected** with logged reasons (*ACP* cloud microphysics and radiation, ozone-only and NO₂-only analyses, water and soil chemistry, life-cycle assessment, and the usual PubMed broad-axis noise). **Five records carried on metadata alone** (38%, a series high) — Elsevier's 5 October deposit is abstract-free across four journals, and the retry leg recovered none. **Consensus** returned 10 calibration hits, **0 new**. **Scholar Gateway** fails with `ACCESS_DENIED`; it has now been unavailable for the whole of this backfill. **ClinicalTrials.gov**: no new PM or air-pollution registration in the window. DOI gate: **0 fail**, **0 warn**. Abstracts remain the copyright of their respective publishers.